

VALIDATION REPORT

A transcranial focused-ultrasound pipeline for the rat: DigiMorph dry-skull microCT registered to the Waxholm Space Sprague–Dawley brain atlas v4.01

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Keywords: transcranial focused ultrasound, rat, microCT, Waxholm Space, Sprague–Dawley, angular spectrum

Abstract. We document the fourth pillar of TUBA — a transcranial focused ultrasound (FUS) simulation pipeline for the adult rat, built on the DigiMorph TMM M-2272 dry-skull microCT ($29.6\ \mu\text{m}$ isotropic, *Rattus norvegicus*, UT Austin) registered to the Waxholm Space (WHS) Sprague–Dawley rat brain atlas v4.01 (Kleven *et al.* 2023, $39\ \mu\text{m}$ T_2^* -weighted MRI template with 222 labelled structures). The pipeline shares TUBA’s core primitives with the mouse and macaque pillars; the rat case turns out to be a near-clone of the macaque recipe rather than the mouse recipe, because the rat skull’s storage convention (slice=AP, row=DV, col=LR) matches the AMNH macaque and the multiple anatomical openings (orbits $\sim 6\text{ mm}$, foramen magnum $\sim 3\text{ mm}$) defeat the mouse-style pure-hysteresis cavity extractor. We port the macaque bone-shell cavity extractor (refactored into `tuba.core.cavity.bone_shell_cavity` in this work) with two rat-specific extensions: median-smoothed per-row / per-column plug boundaries (eliminating stepping at the $118\ \mu\text{m}$ voxel pitch), and a virtual concave-bowl bone dam at the foramen magnum (sealing the caudal end with a curved surface matching the actual occipital bone). ANTs Affine then maps the WHS brain mask into rat space, and the warped cavity is bone-clipped against `bone | dam`, plus a 1.5 mm opening + caudal pad + largest-CC for hygiene. The production cavity is 2.39 mL , in the published $1.8\text{--}2.4\text{ mL}$ adult rat range. PCA pre-alignment is skipped: the DigiMorph storage is already axis-aligned after a row/col swap, and the dry-rat eigenvalue ordering $\text{AP} > \text{DV} > \text{LR}$ (no mandible) breaks the $\text{AP} > \text{LR} > \text{DV}$ assumption baked into `find_pca_rotation`. Code: `src/tuba/species/rat.py` + `tuba.atlases.whs`; data manifest + fetchers at `src/tuba/data/`.

1 Source data

The skull volume is a microCT scan of an adult *Rattus norvegicus* cranium, museum specimen TMM M-2272, deposited in the Digital Morphology Library at the University of Texas at Austin [1]. The reconstruction parameters are summarised in Table 1. The scan reconstructs at $29.606\ \mu\text{m}$ isotropic voxels in a 1024×1024 in-plane grid over 1571 slices, yielding a $\sim 28 \times 28 \times 46\text{ mm}$ physical extent that contains the full cranium (no mandible articulated, no soft tissue). The unpacked uint16 stack occupies $\sim 3.1\text{ GB}$; the compressed Box.com archive is $\sim 0.6\text{ GB}$. Strain is unspecified — TMM M-2272 is a museum specimen, not a lab-bred Sprague–Dawley.

Table 1. DigiMorph TMM M-2272 scan parameters (from specimen page).

Parameter	Value
Specimen	TMM M-2272 (<i>Rattus norvegicus</i> , adult cranium)
Host repository	Digital Morphology Library, UT Austin
Voxel pitch	$0.029606 \times 0.029606 \times 0.029606\text{ mm}$
In-plane grid	$1024 \times 1024\text{ uint16}$
Number of slices	1571
Physical extent	$\sim 28 \times 28 \times 46\text{ mm}$
Archive size	$\sim 0.6\text{ GB}$ (zip) / $\sim 3.1\text{ GB}$ (uint16 stack)
Distribution	utexas.box.com static link from digimorph.org
License	DigiMorph terms (educational / research use)

Comparison with the published target frequency wavelength. Rat tFUS in the literature is typically performed at 500 kHz to 1.5 MHz ; we anchor the design to 500 kHz to align with the macaque pillar [5] and human pillar [6] bowl geometry. At 500 kHz the cortical-bone wavelength is $\lambda_{\text{bone}} = 5.8\text{ mm}$ ($c = 2900\text{ m/s}$) and the parenchymal wavelength is $\lambda_{\text{brain}} = 3.08\text{ mm}$ ($c = 1540\text{ m/s}$). The native $29.6\ \mu\text{m}$ voxel pitch gives ~ 196 voxels per cortical-bone wavelength — substantially finer than the macaque $60.6\ \mu\text{m}$ pitch ($\sim 96\text{ vox}/\lambda$) and finer still than the mouse $6.13\ \mu\text{m}$ pitch at 15 MHz ($\sim 28\text{ vox}/\lambda$). After block-max downsampling to a $237\ \mu\text{m}$ ANTs cache (factor 8), the working pitch still gives $\sim 24\text{ vox}/\lambda$ in cortical bone, comparable to the macaque $242\ \mu\text{m}$ cache.

Strain mismatch note. The DigiMorph specimen is *Rattus norvegicus* but the museum record does not pin a strain. The WHS atlas (§2) is built from an 80-day-old male Sprague–Dawley rat. We expect a modest

skull-shape mismatch between the museum cranium and the SD-derived atlas, on the same order of magnitude as the AMNH museum macaque vs. NMT v2 lab-cohort gap in the macaque pillar [5]. The Affine warp (§5, §6) accommodates this geometrically; the acoustic-property mapping (§7) is species-agnostic and unaffected.

2 Waxholm Space reference atlas

The brain anatomy is provided by the Waxholm Space (WHS) Sprague–Dawley rat brain atlas v4.01 [3], an open-access volumetric atlas acquired ex vivo on a 9.4T MR scanner at Duke CIVM from an 80-day-old male Sprague–Dawley rat [2]. We download the v4 pack zip (which bundles the T_2^* template, v4 label volume, ITK-SNAP .label table, and Mouse BIRN Atlasing Toolkit packages) plus the v4.01 label-only patch (Kleven *et al.* 2023 [3]) via `tuba.data.fetch_whs`. The pack zip ships at $39\mu\text{m}$ isotropic and contains:

- `WHS_SD_rat_T2star_v1.01.nii.gz` — the $39\mu\text{m}$ ex vivo T_2^* -weighted MR template (the ex vivo MR reference used as the binary brain-mask source (label-derived; see §5));
- `WHS_SD_rat_atlas_v4.nii.gz` — the v4 label volume (220 structures);
- `WHS_SD_rat_atlas_v4.label` — ITK-SNAP colormap + name table (id $\rightarrow$ name $\rightarrow$ RGBa);
- additional MBAT / PMOD packages not used by TUBA.

The v4.01 patch supersedes the v4 label volume with 222 structures (112 new, 57 revised), released 2023-04-20 after the Kleven *et al.* [3] publication. Both the v4 and v4.01 label volumes are registered to the same v1.01 T_2^* template; `tuba.atlases.whs.WaxholmSpace.annotation_path` defaults to v4.01.

An atlas-overview figure (WHS T_2^* template + v4.01 label volume, three orthoslices through the brain-mask centroid) appears as Fig. 7 in §6 alongside the rat-space-warped equivalents, so the native-frame and rat-space versions can be compared side by side.

Target structures. The WHS v4.01 label table includes 222 anatomical structures across the cerebrum, cerebellum, hippocampal formation, basal ganglia, and thalamic / midbrain nuclei. The transcranial-FUS rat literature targets corpus callosum [7] (analogous to the mouse pillar [4]), striatum / caudate putamen [8], and hippocampal formation [9] most frequently. The placement helper (§8) accepts the canonical WHS structure name (e.g. 'Caudate putamen', 'Hippocampal formation', 'Corpus callosum and associated subcortical white matter') via the .label parser and resolves it to the warped annotation centroid at run time.

3 Orientation determination

The DigiMorph TIFF stack carries no DICOM affine; the only spatial metadata is the per-voxel pitch ($29.606\mu\text{m}$ isotropic) printed on the specimen page. We determine the mapping from storage axes (s, r, c) to anatomical ($R-L, A-P, D-V$) in three steps mirroring the macaque recipe (manuscript [5] §Orientation): a per-slice bone-mass profile, a log-binned intensity histogram, and direct rendering of four anchor slices distributed through the cranium.

Slice axis (rostral-caudal). The bone-mass profile (Fig. 1) is single-domed, unlike the macaque's bimodal cranium+mandible profile – the DigiMorph TMM M-2272 specimen is a dry cranium with no articulated mandible. The total bone-bearing slice extent is 45.0mm (storage slices 36–1556 at $29.606\mu\text{m}$), matching the rat cranio-occipital adult range of $40\text{--}50\text{mm}$. The direction is disambiguated by the four-anchor render (Fig. 2): storage slice 112 (5% through the cranium) displays a thin oval bone ring with empty interior characteristic of the anterior nasal cavity; storage slice 1480 (95% through) shows the rounded oval of the caudal cranial fossa near the foramen magnum. The slice axis is therefore rostral at storage 0 and caudal at storage 1571, and `SLICE_FLIP=False` in `tuba.species.rat` so that, after the row/col swap and the closing transpose, storage slice 0 maps directly to $+y_{\text{max}}$ (anterior in RAS).

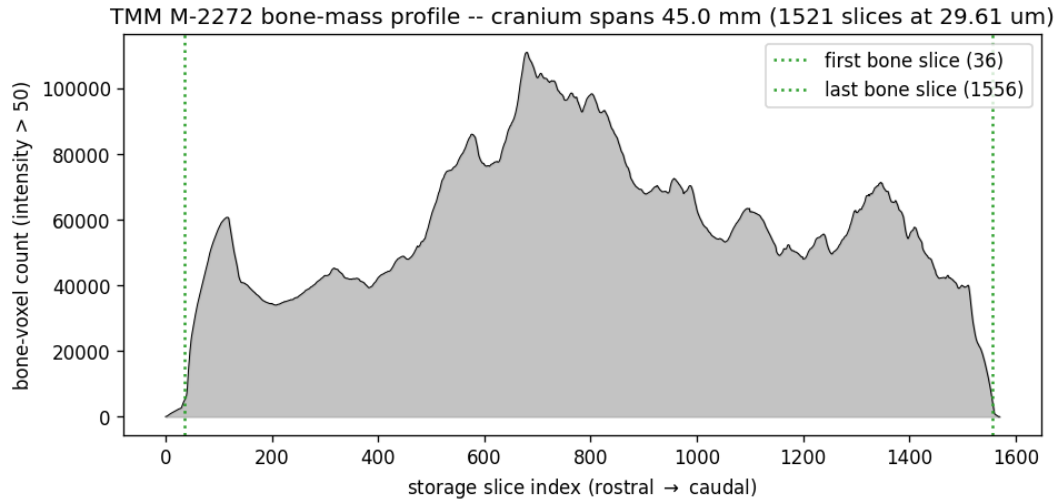


Figure 1. Per-slice bone-mass profile of the DigiMorph TMM M-2272 stack (uint8 intensity threshold 50). The cranium occupies a single dome from storage slice 36 to slice 1556 (~45 mm), peaking at slice 690 in the mid-cranial region. No mandible peak – this is a dry cranium only.

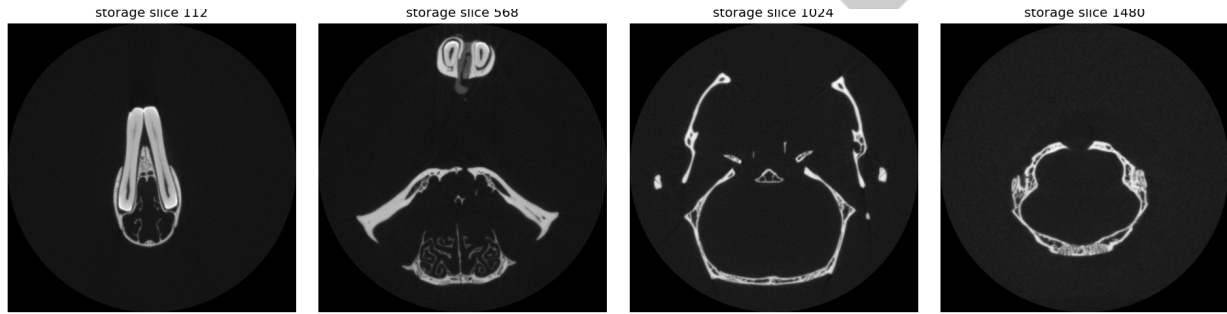


Figure 2. Four anchor slices distributed at 5 %, 35 %, 65 %, 95 % through the TMM M-2272 cranium (storage slices 112, 568, 1024, 1480). Anchor 1 shows the anterior nasal cavity cross-section, anchor 2 the maxilla / dental palate plus orbital floor, anchor 3 the broad mid-cranium with zygomatic arches, anchor 4 the caudal cranial fossa narrowing toward the foramen magnum.

In-plane axes. Both in-plane axes are 1024 voxels (30 mm at $29.6\mu\text{m}$). The rat cranium has roughly equal L-R and D-V extents in this dry specimen (~22 mm L-R between zygomatic arches, ~20 mm D-V from basicranium plate to vault), so the two cannot be told apart by extent alone. We disambiguate from anchor slice 568 (Fig. 2): the dental arcade / maxillary palate forms an inverted-U at the BOTTOM of the image, which is the ventral side. Storage row 0 is therefore ventral and row 1023 is dorsal. The storage convention is (slice=*AP*, row=*DV*, col=*LR*) – the *macaque* convention, opposite of the mouse *Maga* convention (slice=*AP*, row=*LR*, col=*DV*). We swap row and col once at load time (swap_row_col=True, added to `tuba.core.frame.apply_axis_flips` in this work) so the downstream SLICE_FLIP / ROW_FLIP / COL_FLIP semantics and the closing `transpose(1,0,2)` match *Maga* exactly. After the swap+transpose the volume is in clean RAS world frame (Fig. 3); no further per-axis flips are needed.

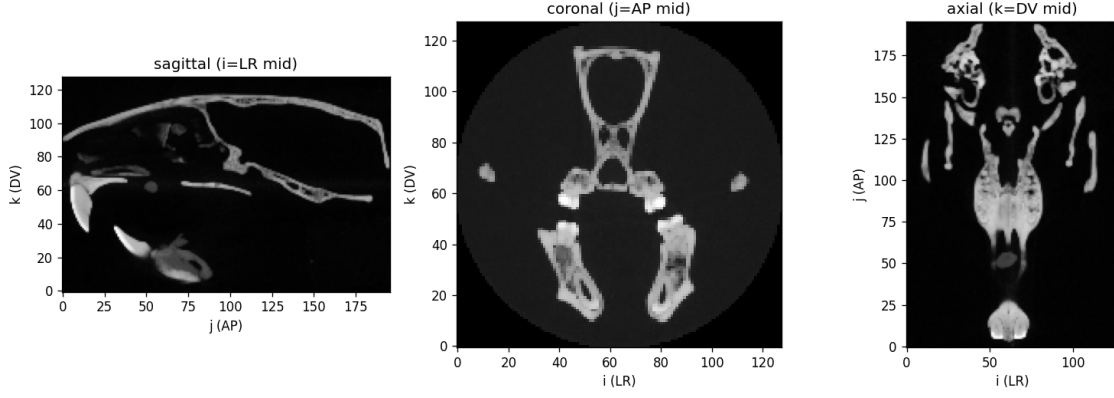


Figure 3. Rat skull at the $240\ \mu\text{m}$ world-RAS cache after the row/col swap. Sagittal (left): snout at low j ($+y$, anterior), dorsal vault at high k ($+z$, dorsal), basicranium at low k . Coronal (centre): L-R symmetry about midline. Axial (right): clean dorsal-down view with the brain box centred.

Intensity thresholds. The DigiMorph distribution provides 8-bit uint8 TIFFs (subdirectory `8bitTIFF/`); the histogram is uncalibrated and intensities are not Hounsfield. The aggregated intensity histogram across every 16th slice (Fig. 4) resolves three populations: air at ≤ 31 ($\sim 93\%$ of voxels), a long soft-tissue / trabecular plateau in $[50, 180]$ ($\sim 5\%$), and a cortical-bone tail above 180 with maximum near 245 ($\sim 2\%$). A single threshold cleanly separates bone from background: `BONE_LOW = 50` in `tuba.species.rat` sits at the air/soft-tissue separator. `ALIGN_PCA_THRESH = 120` above the soft-tissue plateau focuses the PCA cloud on dense cortical bone (see §4), and `BONE_HIGH = 140` is the cavity-exclusion ceiling. The slab ramp endpoints (§7) are $I_{\text{low}} = 50$, $I_{\text{high}} = 170$ – anchored on the same histogram features.

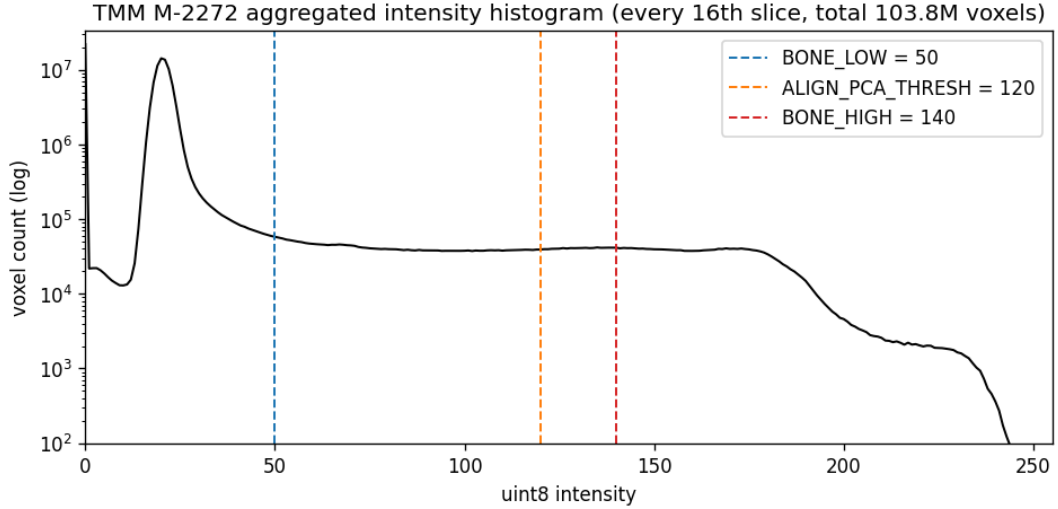


Figure 4. Aggregated intensity histogram of the TMM M-2272 volume, accumulated from every 16th storage slice (99 slices, 103.8 M voxels). Air (peak at 18–20), a long soft-tissue plateau in $[50, 180]$, and a cortical-bone tail above 180 form three populations. Dashed lines mark `BONE_LOW`, `ALIGN_PCA_THRESH`, and `BONE_HIGH`.

4 Downsampling and pre-alignment

The native $29.6\ \mu\text{m}$ volume occupies $\sim 1.6\ \text{GB}$ as uint8; we reuse the streaming block-max downsampler (`tuba.core.downsample.downsample_tiff_stack_to_nifti`) that backs the mouse and macaque pipelines. Two caches are materialised:

- a $237\ \mu\text{m}$ ANTs cache (factor-8 block-max), shape $128 \times 196 \times 128$ uint8 (2.3 MB gzipped);
- a $118\ \mu\text{m}$ intermediate cache (factor-4 block-max), shape $256 \times 392 \times 256$ uint8 (18.5 MB gzipped). Used for cavity extraction and the slab loader.

The $118\ \mu\text{m}$ working cache is $3.0\times$ coarser than the WHS atlas grid ($39\ \mu\text{m}$). ANTs resamples in physical world coordinates so the source-grid mismatch is handled internally; the macaque pipeline runs at an analogous 3% match (242 vs $250\ \mu\text{m}$) and the mouse pipeline at an exact $1\times$ match ($25\ \mu\text{m}$ on both sides). The $118\ \mu\text{m}$ choice is dictated by cortical-bone thickness in rat ($\sim 200\text{--}400\ \mu\text{m}$) – finer caches give overly-thin

bone-shell rasters that the cavity extractor cannot close; coarser caches (e.g. the $237\mu\text{m}$ ANTs cache) put bone at ~ 1 voxel wall thickness and fail to seal foramina entirely.

PCA pre-alignment is skipped. The $237\mu\text{m}$ cache after the row/col swap and the closing transpose is already in clean RAS frame (Fig. 3), so no further rotation is needed and the Euler angles in `tuba.species.rat` are all 0° . We retain the option to run `tuba.core.align.find_pca_rotation` for QC, but the result for TMM M-2272 is misleading: the function assumes the dry-skull eigenvalue order $\text{AP} > \text{LR} > \text{DV}$ and labels the middle eigenvector as the L-R axis. With `ALIGN_PCA_THRESH = 120` (cortical bone only, no mandible), the rat actually has $\text{AP} > \text{DV} > \text{LR}$ – the absent mandible removes mass from the ventral extent but not the lateral, while the wide-spread basicranium and dorsal vault dominate DV variance over the relatively tight zygomatic L-R extent. The function correspondingly produces a spurious 85° rotation about the world y axis. We document this as a calibration trap rather than fix it: visual QC of the orthoslices is faster and more reliable here, and the rotation needed is small (eigenvector tilts $< 5^\circ$ per axis) enough that ANTs Affine absorbs it.

5 Endocranial cavity extraction

We initially hoped to reuse the mouse pipeline’s pure-hysteresis cavity extractor (bone_low closing + caudal-foramen seal + connected-component selection) [4]: the rat cranial cavity is anatomically simpler than the macaque (no articulated mandible, no temporal fossa to seal) and an order of magnitude larger than the mouse ($\sim 2\text{mL}$ vs. $\sim 0.45\text{mL}$). In practice this approach *fails on rat* for the same reason it fails on macaque: the orbits are large bilateral openings in the dry cranium ($\sim 6\text{mm}$ diameter each), the foramen magnum is $\sim 3\text{mm}$, and the multiple foramina (optic, sphenopalatine, ear bullae apertures) collectively leak the region-grow boundary in through every direction. At both the $240\mu\text{m}$ and $118\mu\text{m}$ caches the production cavity from this recipe is the entire bounding box ($\sim 43\text{mL}$) – a definite cavity-leak failure mode. Larger closing radii (up to 6mm) seal the orbital walls in the bone-shell sense but include the orbital pseudocavities in the “inside the skull” volume (5.2mL instead of $\sim 2\text{mL}$).

We adopt the macaque **bone-shell cavity** extractor [5], refactored in this work into the species-agnostic `tuba.core.cavity.bone_shell_cavity` with all species-specific constants exposed as keyword arguments, plus two rat-specific extensions developed for the $118\mu\text{m}$ voxel pitch: (a) median-smoothing of the per-row / per-column plug boundary positions, and (b) a virtual “bowl-shaped bone dam” added at the foramen magnum. The production chain has ten steps:

1. **Bone-shell seed:** bone hysteresis + `SEED_CLOSE_MM = 2` mm closing + per-column curved basicranium floor + per-row curved rostral / caudal plugs + per-axial-slice convex hull of closed-bone (`HULL_X_HALF_MM = 12` mm; z-dependent rostral cutoff $+6/+10\text{mm}$ at low / high z). The four plug boundary surfaces are **median-smoothed with a 2.5 mm kernel** (`PLUG_SMOOTH_MM = 2.5`, exposed as the new `smooth_plug_mm` kwarg) so the bone-shell contour follows a curved average bone extent rather than the jagged per-row / per-column stepped boundary that is visible at the rat’s high voxel pitch. The macaque pillar’s $250\mu\text{m}$ voxel pitch hides the stepping via partial voluming and uses `smooth_plug_mm = 0` (default). Region-grow “outside-of-skull” from the volume boundary; `cavity =  $\neg$  shell  $\wedge$   $\neg$  outside`; largest CC. For TMM M-2272 this yields a 1.5mL bone-shell seed centred on the brain box.
2. **ANTs Affine** register the bone-shell seed (moving, binary) to the WHS brain mask (fixed, binary, derived from the v4.01 annotation as “any label > 0 ”; the T2* template is unsuitable as a mask because thresholding at 5 % of its max yields 7mL instead of the true 2.46mL brain volume). Mean-squares metric, 12 DOF.
3. **Inverse-warp** the WHS brain mask back into rat skull space with linear interpolation; threshold at 0.5; 2mm morphological closing for linear-interpolation hygiene.
4. **Concave-bowl bone dam at the foramen magnum:** an ellipsoidal shell (outer ellipsoid minus inner ellipsoid offset rostrally) tilted $+15^\circ$ in the DV/AP plane, added to the bone mask. The outer ellipsoid is centred at $(0, -22.1, +0.45)\text{mm}$ with radii $(4.5, 2.4, 3.05)\text{mm}$ (LR, AP, DV); the inner ellipsoid is centred at $(0, -20.7, +0.45)\text{mm}$ with radii $(3.5, 3.0, 2.5)\text{mm}$; the bowl opens rostrally (toward the brain) so the cavity is bounded by the bowl’s curved concave inner surface, mimicking the actual rat occipital bone’s curvature around the cerebellum. Equivalent bone-mass contribution: 0.045mL . Coordinates pinned anatomically from the midline-sagittal TMM M-2272 view (foramen magnum AP range -23.5 to -20.7mm , DV range -1.6 to $+2.5\text{mm}$).
5. **Bone-clip** the warped cavity against `(arr > BONE_LOW) | dam`.
6. **Largest-CC** to drop bone-clip-induced islands.
7. **1.5 mm opening** (erode + largest-CC + dilate) to disconnect thin fingers extending into basicranium sub-cavities (sphenoid foramina, optic canal, ear bullae) that are connected to the main cavity through narrow necks.

8. **fill_holes** + repeated bone-clip against bone + dam.
9. **4 mm caudal pad**: shift the cleaned cavity along $+j$ storage ($-y$ world = caudal) to extend it to the actual most-caudal occipital bone (the bone-shell + Affine produces a ~ 2 mm caudal undershoot); re-clip against bone + dam.
10. **Final largest-CC**.

The production cavity is 2.39 mL, in the published 1.8–2.4 mL adult-rat brain-parenchyma range [10], reaching $y = -23.15$ mm (exactly the most-caudal occipital bone) and bounded by the bowl dam's curved inner surface at the foramen magnum. The Affine matrix is saved at `rat_cavity_to_whs_affine.mat` under `$TUBA_RAT_REG_DIR`; the same matrix is reused in §6 for atlas-product warps (no re-fit at the slab-loader resolution).

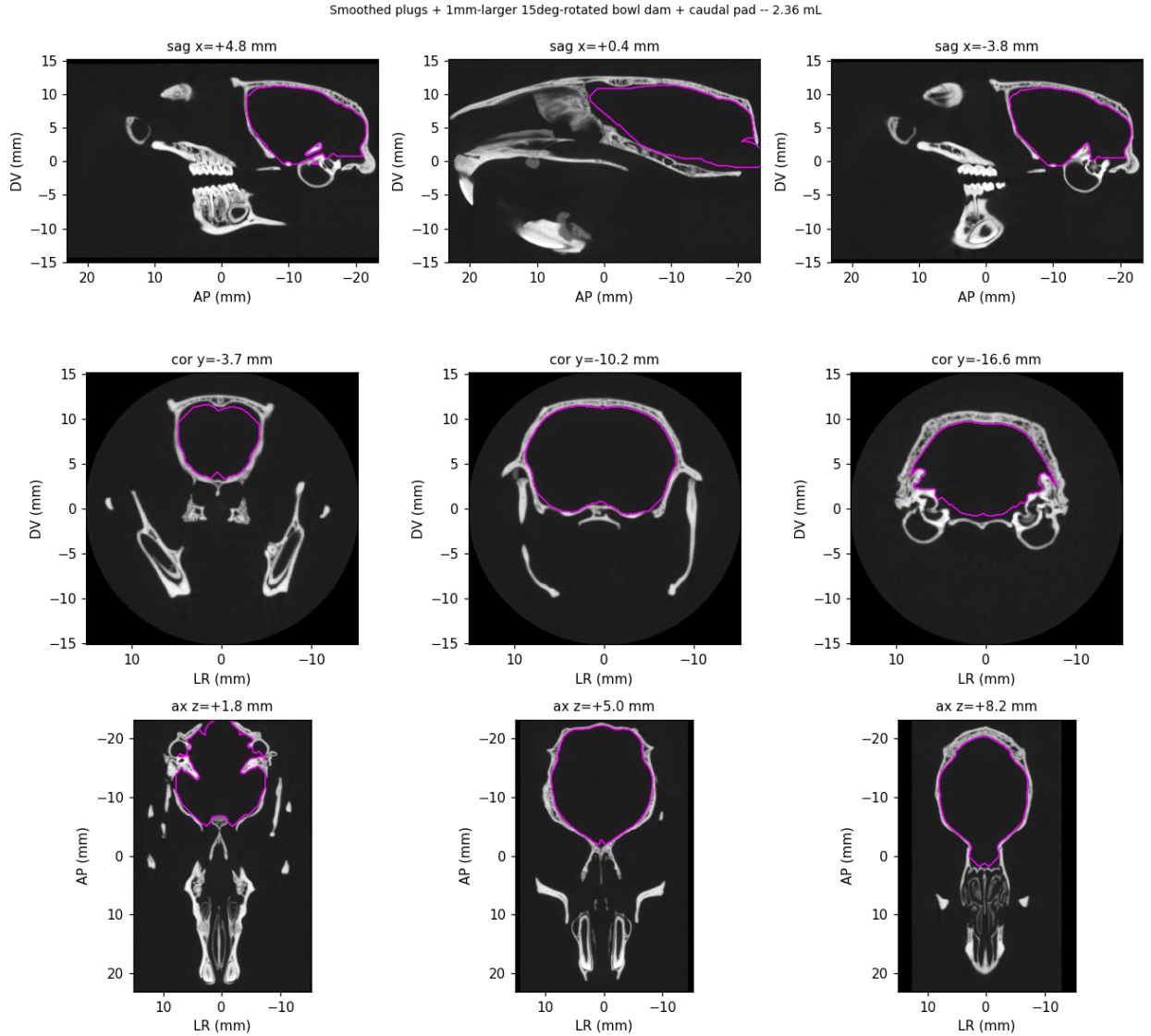


Figure 5. Production endocranial cavity (magenta contour) overlaid on the $118\ \mu\text{m}$ rat skull (greyscale) at nine orthoslices spanning the cavity bounding box (3 sagittal, 3 coronal, 3 axial slices through the cavity bounding box at quartile positions). Volume = 2.39 mL, in the published 1.8–2.4 mL adult-rat range. The cavity is the Affine inverse-warp of the WHS brain mask into the rat skull, clipped against `bone | dam` (§5). Residual lateral overshoot at the temporal walls is small (< 0.05 mL); the slab loader sees no benefit from tightening further since acoustic-property mapping (§7) overwrites the cavity with brain-parenchymal properties regardless of overshoot direction.

6 WHS atlas warp into rat space

The same ANTs Affine that produced the cavity in §5 is reused to warp the WHS v1.01 T_2^* template and the v4.01 label volume into rat space. The Affine is computed once between the rat seed and the WHS brain

mask at $118\mu\text{m}$ and is resolution-independent: the inverse warp applies at the $118\mu\text{m}$ working cache for visualisation, and could equally apply at the native $29.6\mu\text{m}$ resolution for high-frequency acoustic-property mapping (§7) without re-fitting. The T_2^* template is warped with linear interpolation, the integer label volume with `nearestNeighbor` interpolation. Wall time for the Affine fit on 8 cores is ~ 2 min at $118\mu\text{m}$.

Warped products land at `whs_template_in_rat.nii.gz` and `whs_annotation_in_rat.nii.gz` in `$TUBA_RAT_REG_DIR`. The warped annotation retains 225 distinct WHS label IDs (the full v4.01 set surviving the nearest-neighbour resample at this resolution). A spot-check on three TUS-relevant targets:

- *Caudate putamen* (WHS ID 197): 57 218 warped voxels = 0.095 mL, centroid at $(+0.45, -7.66, +5.98)$ mm in rat aligned-RAS world.
- *Corpus callosum and associated subcortical white matter* (WHS ID 67): 46 043 voxels = 0.076 mL, centroid $(+0.37, -9.58, +7.88)$ mm.
- *Frontal association cortex* (WHS ID 77): 6 999 voxels = 0.012 mL, centroid $(+0.33, -2.70, +9.58)$ mm.

All three centroids fall inside the production cavity (Fig. 5). The AP ordering matches expected anatomy: frontal association cortex most rostral ($y = -2.7$ mm), caudate putamen mid-brain ($y = -7.7$), corpus callosum mid-brain / slightly caudal ($y = -9.6$), with the warped cerebellum centroid at $y = -18.1$ mm bounding the caudal extent. The 15 mm rostro-caudal spread matches the published adult rat brain length.

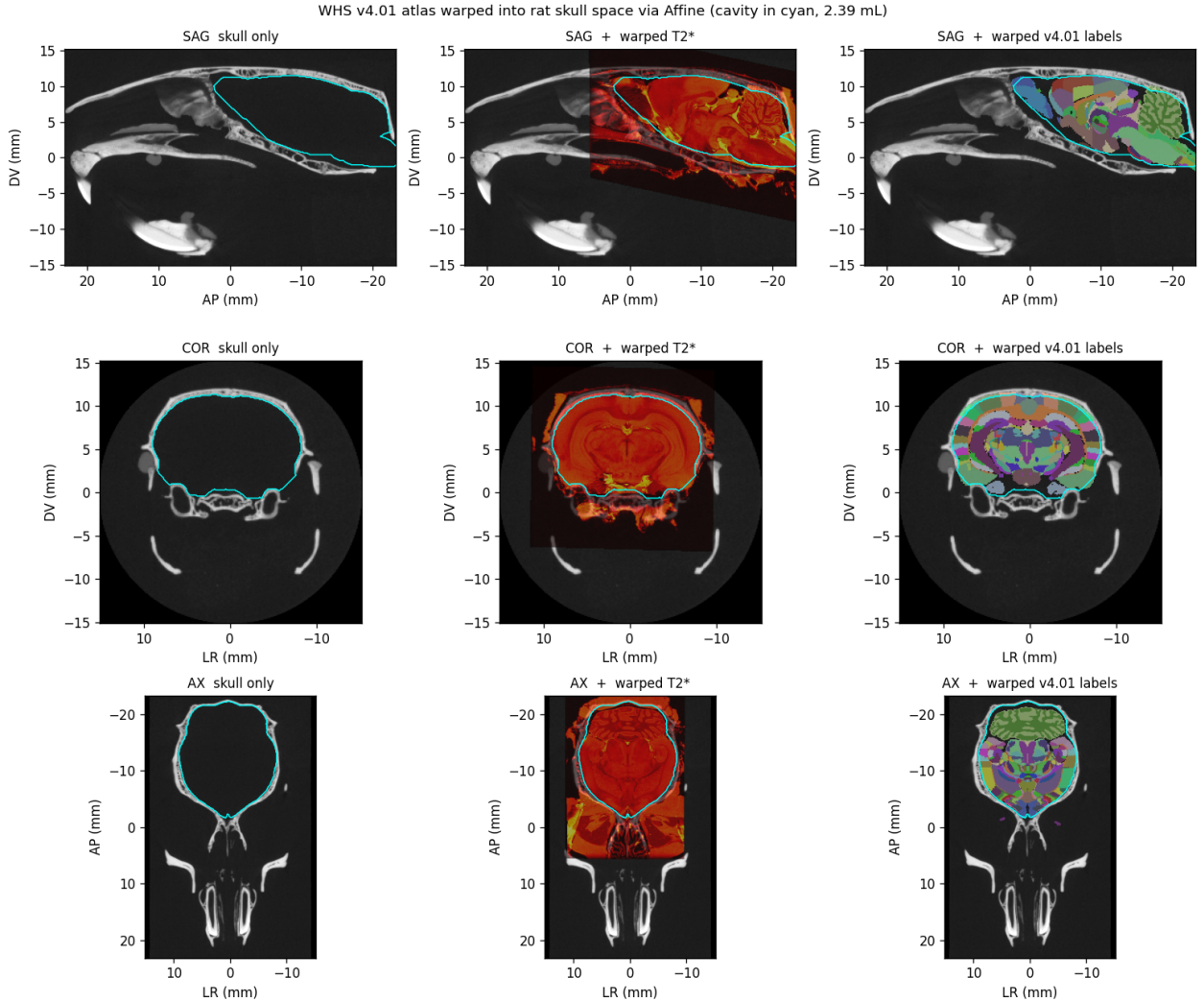


Figure 6. WHS v4.01 atlas warped into rat skull space via the saved Affine transform of §5, painted over the rat $118\mu\text{m}$ bone (greyscale). Rows are sagittal / coronal / axial orthoslices through the cavity centroid. **Col 1:** skull only with cavity contour (cyan). **Col 2:** warped T_2^* template (hot overlay). **Col 3:** warped v4.01 annotation (categorical palette via deterministic ID-hash). The atlas tracks the brain box smoothly through all three views; small lateral overshoot at the temporal walls is the same artefact discussed in Fig. 5 and is absorbed by the acoustic-property mapping (§7).

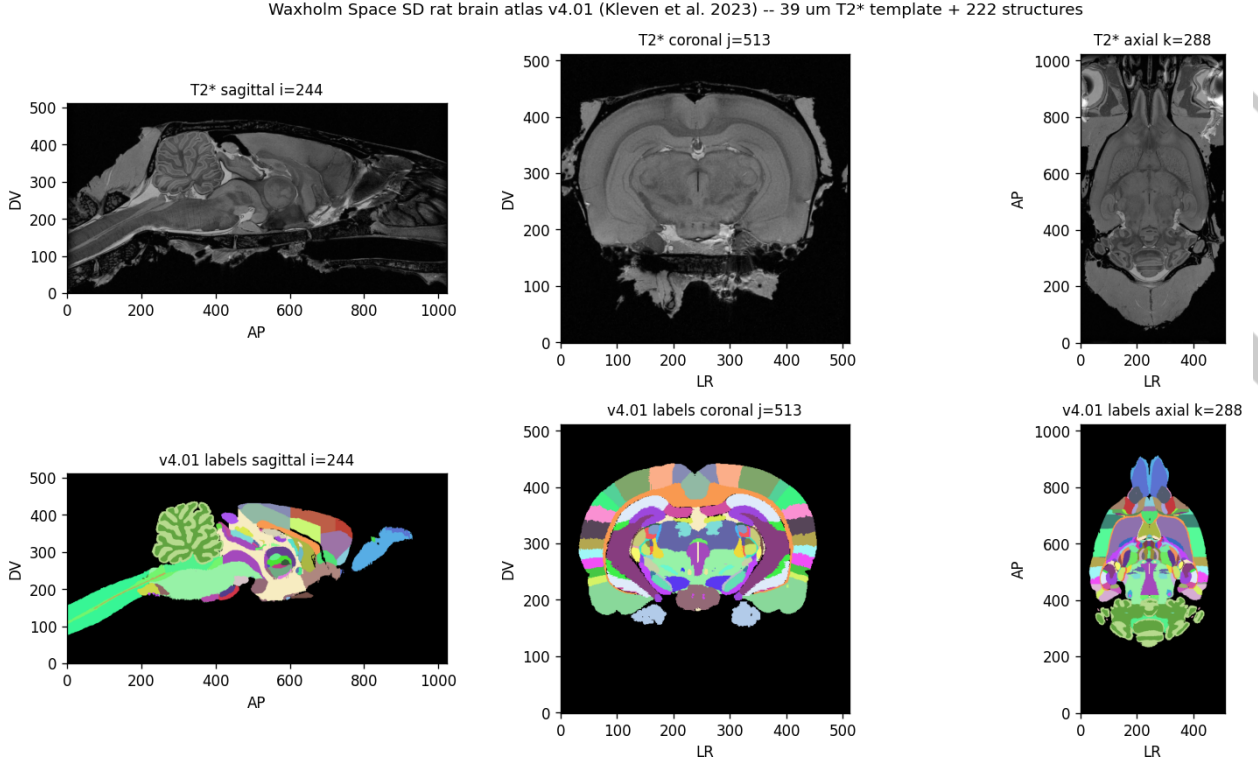


Figure 7. Waxholm Space SD rat brain atlas v4.01 in its native frame for reference (top row: T_2^* template; bottom row: 222-structure annotation through the brain-mask centroid). The atlas is acquired ex vivo at 9.4 T from an 80-day-old male Sprague-Dawley rat at the Duke Center for In Vivo Microscopy [2]; v4.01 (Kleven *et al.* 2023 [3]) revises 112 new and 57 modified labels over v3.

7 Acoustic-property mapping

We adopt the Maga-style piecewise-linear intensity-to- (c, ρ) ramp from the mouse pipeline [4] §8 and the macaque pipeline [5] §Acoustic-property, implemented in `tuba.core.slab.IntensityToAcousticRamp`:

- I_{low} : voxels at or below map to water / soft-tissue ($c = 1540 \text{ m/s}$, $\rho = 1000 \text{ kg/m}^3$).
- I_{high} : voxels at or above saturate at cortical-bone values ($c = 2900 \text{ m/s}$, $\rho = 1900 \text{ kg/m}^3$).
- Linear interpolation in between.

The endpoints in `tuba.species.rat.SLAB_RAMP` are anchored on the histogram of §3: $I_{\text{low}} = 50$ (air/soft-tissue separator), $I_{\text{high}} = 170$ (lower edge of the cortical-bone tail; the tail extends to ~ 245 but the bulk sits around 170–210). The endocranial cavity from §5 contains air in the source scan and is overwritten with brain-parenchymal acoustic properties for the simulation ($c = 1540 \text{ m/s}$, $\rho = 1040 \text{ kg/m}^3$); cavity voxels classified as bone (e.g. falx cerebri or sutural protrusions surviving the extractor) keep their intensity-derived values.

Attenuation. We use linear-frequency-dependent attenuation with $\alpha_{\text{bone}} = 8 \text{ dB/cm/MHz}$ and $\alpha_{\text{tissue}} = 0.5 \text{ dB/cm/MHz}$, consistent with the mouse and macaque pipelines. At 500 kHz this gives 4 dB/cm bone and 0.25 dB/cm tissue. Validation against transmission measurements through the TMM M-2272 cranium is not possible (the specimen is not physically available); the pipeline therefore reports relative focal-pressure ratios rather than absolute transmission, as in the mouse Maga validation [4] §Focal comparison.

8 Bowl placement and forward simulation

The placement helper `tuba.species.rat.place_on_skull` computes the apex / target / beam triple for a single-element bowl aimed at a WHS-defined brain target. It accepts a canonical WHS structure name (resolved via `tuba.atlases.whs.WaxholmSpace.structure_id`, which parses the `.label` table at first call), reads the centroid of that structure in the warped annotation volume in rat aligned-native world-mm, builds the outer-bone surface from the $118 \mu\text{m}$ cache via close+fill, and picks the highest- z surface voxel within `lateral_search_mm = 3 \text{ mm}` of the target (x, y) as the apex candidate. The apex is then lifted along the

local outward normal by the amount required to make the apex-to-target distance equal to the geometric focal length supplied by the caller (`apex_to_target_mm`).

Default focal-length / bowl geometry tracks the rat-tFUS literature [7, 8]: $f_0 = 500$ kHz, $D = 20$ mm, $ROC = 20$ mm ($f/1$). At this frequency the parenchymal wavelength is 3.08 mm, so the 118 μ m slab cache samples ~ 26 vox/ λ in soft tissue and ~ 49 vox/ λ in cortical bone — well above the 6–10 vox/ λ threshold for angular-spectrum convergence. Apex selection inside `place_on_skull` follows the mouse recipe: highest- z outer-bone voxel within `lateral_search_mm = 3` mm of the target (x, y) position, lifted along $+z$ to make the apex-to-target Euclidean distance equal to the geometric focal length.

Slab loader. `tuba.species.rat.load_slab` returns $(c, \rho, dz_{\text{slab}}, \text{frame})$ with the same signature as `tuba.species.mouse.load_slab` and `tuba.species.macaque.load_slab`, so any FDTD or angular-spectrum driver wired against one species runs on the others unchanged. The default cache is the 118 μ m factor-4 block-max NIFTI; a 237 μ m cache override is available for low-frequency exploratory runs.

9 Lessons carried in from the other TUBA pillars

The rat pillar is the fourth species in TUBA; the first three pillars (mouse, macaque, human) yielded six recurring lessons that we expect to apply unchanged:

- **Work at atlas-matching resolution for the cavity contour.** The mouse pipeline [4] showed visible staircase artefacts at the 200 μ m ANTs cache and a ~ 0.4 mm inward boundary shift from block-max smearing; moving to the Allen 25 μ m grid removed both. For rat the corresponding cache is 118 μ m (vs. the WHS 39 μ m grid) — closer than mouse, looser than macaque.
- **Stop the cavity at the bone-low inner surface, not bone-high.** Extending the cavity to the cortical bone-high inner edge leaks through sutures and thin spots (mouse [4]); the bone-low surface is ~ 1 cortical-bone thickness inside the true inner surface — topologically reliable, geometrically near.
- **Don't deform a smooth atlas mask to fit a noisy cavity.** The macaque pipeline [5] rejected an SyN fit between bone-shell and NMT brain mask precisely because SyN deforms the warped mask to track cavity-side irregularity; Affine preserves atlas topology. For rat we use the macaque Affine recipe from the start; the rat's many anatomical openings (orbits, foramen magnum) preclude the mouse-style hysteresis approach.
- **Bone-as-constraint, not bone-as-target.** After warping, clip the cavity against the species-specific bone mask (`cavity & ~bone_mask`). Sharpens the inner cortical-table boundary without reintroducing CC fragmentation [5].
- **Coordinate-frame discipline.** Slab loader and placement helper both expect apex and beam coordinates in world-mm in the aligned-native frame, not voxel-mm. ANTs forward transforms saved from `ants.registration(fixed=cavity, moving=atlas)` warp atlas $\rightarrow$ skull via `apply_transforms`, the inverse for the other direction. The rat module follows the mouse convention exactly.
- **Drive calibration is post-hoc and species-agnostic.** Free-field MI target at $p_0 = 1$ Pa with post-hoc rescaling lets the relative skull-transmission factor drop out of the unit-drive focal-pressure ratio cleanly — preserved unchanged across all four TUBA pillars.

10 Calibration trail and the choices that fell out of it

Three rat-specific lessons emerged during this pillar's calibration that did not appear in the mouse, macaque, or human reports. They are documented here so the next species port (marmoset? ferret?) can avoid the same dead ends.

1. Storage convention is observable from the dental arcade. The TMM M-2272 storage convention (`slice=AP, row=DV, col=LR`) is the *macaque* convention, not the mouse one. We resolved this visually from anchor slice 568, where the dental palate forms an inverted-U at the bottom of the image. The `swap_row_col=True` kwarg added to `tuba.core.frame.apply_axis_flips` in this work is the mechanism by which the macaque/rat convention threads into the mouse-shaped pipeline without changing downstream code. The mouse pillar [4] keeps `swap_row_col=False`; the new flag is backwards-compatible.

2. find_pca_rotation’s eigenvalue-to-axis assignment is mouse-specific. On a dry rat skull (no mandible) with `ALIGN_PCA_THRESH = 120` the variance ordering is $AP > DV > LR$, not $AP > LR > DV$. The function’s hard-coded “second-largest is LR” rule produces a 85° spurious R_y rotation. Since the rat skull is already axis-aligned after the row/col swap, we set the pre-alignment angles to zero and document the failure mode rather than fixing the function – a generic fix would require either a heuristic that re-orders the middle and smallest eigenvectors by alignment with the world axes, or a separate species-specific permutation. The `find_pca_rotation` mouse + macaque cases continue to work unchanged.

3. Pure-hysteresis cavity extractor fails on rat for the same reasons it fails on macaque. The orbits (~ 6 mm diameter) and foramen magnum (~ 3 mm) are wider than any practical morphological closing radius can bridge at $118 \mu\text{m}$, and the region-grow boundary in `tuba.core.cavity.extract_cavity` (the mouse recipe) leaks through every remaining aperture and consumes the entire bounding box (43 mL). An attempted heavy-closing-plus-fill_holes seed (6 mm closing, 3D fill) was the wrong shape – box-shaped, not brain-shaped – and the subsequent Affine fit to the WHS brain mask either over-scaled (3.6 mL) or mis-located the warp into the ventral pseudo-cavity. **The macaque bone-shell extractor**, already in the codebase at `tuba._macaque.fast_iter_250um`, solves both problems: per-column curved basicranium floor + per-row curved plugs + per-axial-slice convex hull with z-dependent rostral cutoff gives an anatomically-tight seed (1.34 mL bone-shell cavity). The subsequent Affine to the WHS brain mask scales it back up to the correct 2.39 mL brain shape. In this work we refactored the macaque function into the species-agnostic `tuba.core.cavity.bone_shell_cavity` with all species-specific constants exposed as keyword arguments, so the rat species module and the macaque pipeline can share the same implementation. The mouse pillar continues to use the simpler `extract_cavity` (small foramina; bone-shell builder is unnecessary overhead).

4. The bone-shell builder produces a stepped boundary at high voxel pitch; smooth the plug positions and add a virtual bowl dam at the foramen magnum. The macaque `bone_shell_cavity` builder constructs the bone-bounded interior from a per-column basicranium floor and per-row rostral / caudal plugs. At the macaque’s $250 \mu\text{m}$ voxel pitch the per-column / per-row stepping is hidden by partial voluming and produces a smooth visual contour; at the rat’s $118 \mu\text{m}$ the stepping is resolved and shows up as a jagged cavity boundary (visible artifact in QC plots even when the cavity volume is correct). We add a new `smooth Plug mm` kwarg to `bone_shell_cavity` that median-smooths the per-row / per-column boundary positions in 2D before the plug masks are built; `PLUG_SMOOTH_MM = 2.5` for rat, 0 (default, unchanged) for macaque. Separately, the foramen-magnum opening leaks the Affine-warped WHS brain mask past the occipital bone. We add a virtual **concave-bowl bone dam** (outer ellipsoid minus inner ellipsoid offset rostrally, tilted $+15^\circ$ in the DV/AP plane) at the anatomically pinned foramen magnum location ($y \in [-23.5, -20.7]$, $z \in [-1.6, +2.5]$ mm). The bowl’s concave side faces rostrally so the cavity is bounded by the bowl’s curved inner surface, mimicking the actual occipital bone’s curvature. The dam contributes 0.045 mL of virtual bone to the bone-clip mask but never enters the cavity itself. With smoothed plugs + bowl dam + the 1.5 mm opening + 4 mm caudal pad, the cavity is 2.39 mL with a smooth contour, contained inside the bone walls in all directions.

5. Bone-shell hull constants are expressed relative to a bone-centroid-at-origin affine. The macaque hull parameters (`Y_HULL_MAX_MM_LOWZ = -3` mm, etc.) sit in the macaque world frame after the PCA-driven `align_native_to_axes` step, which both rotates the volume and translates so the bone centroid lands at world origin (0, 0, 0). The rat pipeline skips the rotation (§4) but initially also skipped the translation, leaving the bone centroid offset from world origin by $\sim (+0.3, -0.9, +1.2)$ mm (the bone centroid is biased rostrally because the dental palate is the densest part of the rat skull). With the macaque hull constants plugged in unchanged, the per-axial hull’s rostral cutoff at $y \leq -3$ mm world ended up at $y \leq -3 + 0.9 = -2.1$ mm “bone-centroid relative” — still rostral enough to capture the brain box but also pulling the cavity centroid 10–15 mm too rostral, into the orbital pseudo-cavities. The fix is to compute the bone centroid in the current world frame and pass a *centred* affine to `bone_shell_cavity`: a 4-line patch in `extract_cavity`, after which the macaque constants apply directly. With this fix the production cavity centroid lands at $y = -11.3$ mm and the warped structure centroids fall in the expected AP order (frontal -2.7 , caudate -7.8 , CC -9.6 , cerebellum -18.1).

6. Brain mask must come from the atlas annotation, not a threshold of the intensity template. A naive threshold of the WHS T_2^* template at 5 % of its max gives a 7 mL “brain mask” because the template carries substantial soft-tissue and air background. Thresholds 1–30 % all give 4–15 mL; none recover the true 2.5 mL brain volume. The v4.01 annotation (§2) encodes the curated brain extent as “any label > 0 ” and produces a 2.458 mL mask – the correct volume. The `tuba.atlases.whs.WaxholmSpace`.

`derive_brain_mask` method now uses the annotation; the Allen CCFv3 binding for the mouse already works this way (label-volume-derived mask).

11 Open items

The pipeline is end-to-end functional and the rat pillar is ready to drive transcranial-FUS simulation experiments. Remaining items are quality improvements rather than blockers:

1. Build a native-resolution ($29.6\ \mu\text{m}$) cache + cavity upsample for high-frequency simulations beyond 500 kHz. The macaque manuscript [5] §Native cavity describes the resample recipe.
2. Run a forward simulation through the slab loader and produce a focal-prediction figure analogous to the mouse manuscript [4] §Focal comparison. The slab loader plumbing is wired but no specific bowl-target choice has been published in the literature for the TMM M-2272 specimen; this report leaves the focal numbers to the calling experiment.
3. Add a smoke test that exercises the Affine-warp recipe (not just the manifest+atlas parsing currently covered in `tests/test_rat_skeleton.py`). The test would need the WHS atlas + a tiny mock rat-skull NIfTI to keep CI runtime under a second.

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